

2024



lexoffice Scales to Meet Growth, Cuts User Response Times by 50%

A leading business software provider in Germany, [Lexware](#) has overcome the performance and reliability issues of its SaaS accounting software product, lexoffice, by migrating its entire infrastructure to Amazon Web Services (AWS). The company now has more development flexibility and can scale easily to accommodate rapid customer growth. Using AWS, it has reduced response times by 50 percent, can scale to meet monthly usage peaks, and provide information queries almost instantly rather than in several hours. It also uses a range of AWS security services to help it to protect sensitive customer data, made easier because of their tight integration with the AWS services it uses.

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Lexware, based in Germany, is a provider of business software for small businesses and freelancers. One of its market leading software products is lexoffice, an accounting software as a service (SaaS) solution. The lexoffice product team started to experience IT performance issues in 2015 and again in 2019—both of which had a negative impact on the end-user experience. The product team had run its multi-tenant SaaS product on premises from its launch in 2012 to safeguard sensitive customer data and comply with complex German tax accounting rules.

When it first experienced bottlenecks, lexoffice migrated part of its post-production system to Amazon Web Services (AWS). As customer numbers grew and it experienced further scaling issues, the company decided to migrate all its operations to the cloud and worked with AWS to make sure its data would be fully protected and compliant with local tax rules following migration. Running on AWS, the company has cut database response times by 50 percent and customers can now rapidly access accounting information with sub-second data retrieval.



Opportunity | Supporting Development of New Product Features Using AWS

lexoffice, a SaaS product developed by Lexware, is a market leader in Germany for accounting software with 250,000 active daily users that include sole traders, micro-businesses, and small businesses. The product's accounting modules comprise billing, invoicing, payroll, banking, and tax return services—all of which help simplify accounting procedures so customers can focus on developing their business. lexoffice also helps its users to comply with complex German tax reporting regulations—called GoBD—that lay out the IT requirements for the processing and recording of tax-relevant data. The regulations include data security measures to ensure all digital tax records can be protected, traceable, and easily verified.

In 2015, lexoffice began to run into performance and reliability issues that were having a negative effect on the end-user experience. It had been using AWS for testing and development from day one, building this facility on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload.

Because of its previous positive experience with AWS, the lexoffice product team migrated part of its system to AWS to improve product performance, using [Amazon Simple Storage Service](#) (Amazon S3), which is built to retrieve any amount of data from anywhere. The company then began building the majority of all its new



product features as separate microservices directly on AWS, including bespoke invoice templates and a PDF-generation module.

When lexoffice experienced further performance issues in 2019, the company knew it needed to act decisively. Rather than carry out a costly upgrade to its on-premises setup, the company felt that the best move would be to go all-in on AWS. “We needed to be able to improve product performance for existing customers and to more efficiently onboard new ones to keep up with growing market demand,” says Christian Steiger, managing director at Lexware. “We saw that AWS could support our rapid growth and scale to maintain a solid end-user experience.”

Solution | Supporting Scaling and Complying with Local Tax Regulations

lexoffice began its migration by first moving its databases to AWS using [Amazon Aurora](#), which provides unparalleled high performance and availability at global scale with full MySQL and PostgreSQL compatibility. This cut response times by 50 percent. By building on AWS, lexoffice can scale to meet peak usage demands and it can also reduce infrastructure usage when needed, for example during the night when customer demand falls.

Because of the need to protect customers’ confidential financial information, the company uses a range of AWS security and data encryption services. These include [AWS WAF](#)—integrated with [Elastic Load Balancing](#) (ELB)—to protect against common web exploits, [AWS Shield](#) for managed DDoS protection, and [Amazon GuardDuty](#) to continuously monitor for malicious activity. “Another key benefit of AWS is the ability to give us a complete picture of our level of security at any time, as all its services are tightly connected,” says Steiger. “Using AWS services, we can configure the level of encryption we need, giving us peace of mind that our cloud set up keeps our customers’ financial records secure.”

The company also uses AWS to help its customers compile all financial activity and comply with the GoBD tax regulations. Using activity logs, important data can be retrieved almost immediately instead of having to wait hours. For this it uses [Amazon DynamoDB](#), a flexible NoSQL database service for single-digit millisecond performance. “Our customers can easily locate information and provide accurate supporting data for all financial reporting,” says Steiger. “If we didn’t use AWS, we’d need to have a team to do regular performance reviews to see if we needed to increase database capacity—because we can scale automatically, all that goes a



The lexoffice IT team now has more time to focus on developing new products and features, which had been a much slower process with its on-premises setup. The team has saved time on managing servers by adopting the concept of immutable infrastructure. This has delivered increased infrastructure consistency and reliability, with increased predictability in the deployment process. Using AWS serverless services, the system can automatically scale to meet traffic and demand peaks. In addition, lexoffice can now support webhooks, allowing customers to configure application events to automatically trigger events in external software systems, such as when an invoice has been paid or creating new invoices when they need to be issued automatically. For this, it uses [AWS Lambda](#), which allows users to run code in response to events without thinking about servers or clusters.

Using AWS, the company has been able to create a DevOps culture that helps its lexoffice teams handle performance issues much faster. “The IT team no longer has to deal with items such as manual updates, scaling, and monitoring because AWS services have a higher level of abstraction,” says Steiger. “With AWS managed services, we don’t need to set up teams to handle items such as security updates and scaling. AWS tells us about updates and only asks us when they should apply changes—that’s a huge time saver for us.”

Outcome | Expanding Use of AWS to Support Growth

Using AWS, lexoffice now has the flexibility to scale and grow its online customer base without compromising its service. It plans to start moving its development platform from Amazon EC2 to [Amazon Elastic Container Service](#) (Amazon ECS) to run highly secure, reliable, and scalable containers.

The company also plans to evolve its multi-account strategy to enhance and build on the AWS environment. “We are expanding and reaching new customers all the time so it’s really important our IT infrastructure supports that,” says Steiger. “We gained 93,000 new customers in the last year alone and are confident that we can now grow painlessly—something we wouldn’t have been able to do without AWS.”

About lexoffice

lexoffice is online accounting software for the German market, developed by Lexware, serving 250,000 small business, entrepreneur, and freelance customers. The product makes accounting processes easy, allowing customers to focus on business development.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. Access reliable, scalable infrastructure on demand. Scale capacity within minutes with SLA commitment of 99.99% availability.

[Learn more »](#)



Amazon Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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